locality_diagnostics: how much of the trace a model needs

What this folder shows

This check from the long version takes a trained network and shows it only a short window of each trace, with everything outside the window set to zero. For each window length it finds a good window position and records how often the network still guesses right. The paper plots that success rate against the window length.

Before you start

The Docker image (or the Python environment with `requirements-ml.txt`) and the TCHES20 download, which includes the ASCAD traces file and the trained networks. One window length with one seed takes about a minute. The full sweep in the paper (nine lengths, five seeds, a search over positions) takes about an hour.

Run

```
T=datasets/raw/tches20/TCHES20V3_CNN_SCA
python -m locality_diagnostics.ascad_temporal_window_sweep \
  --ascad-h5 $T/datasets/ASCAD_dataset/ASCAD.h5 \
  --cnn-model $T/models/pretrained_models/models/zaid_ascad_desync_0_no_preprocessing_0.hdf5 \
  --tau-values 50,700 --n-seeds 1 --strategy center --output-dir results/locality_diagnostics
```

```
$ T=datasets/raw/tches20/TCHES20V3_CNN_SCA; python -m locality_diagnostics.ascad_temporal_window_sweep --asc
    ad-h5 $T/datasets/ASCAD_dataset/ASCAD.h5 --cnn-model $T/models/pretrained_models/models/zaid_ascad_desyn
    c_0_no_preprocessing_0.hdf5 --tau-values 50,700 --n-seeds 1 --strategy center --output-dir results/local
    ity_diagnostics
[... earlier output omitted; key lines kept ...]
[...]

  Monotonicity check: PASSED
  Random guess: 0.003906

======================================================================
COMPLETE: 2 new pairs computed, 2 total
======================================================================
  Summary saved to: results/locality_diagnostics/tau_sweep_summary.csv
  Saved: results/locality_diagnostics/figures/tau_sweep_bi.pdf
  Saved: results/locality_diagnostics/figures/tau_sweep_success.pdf
  Saved: results/locality_diagnostics/figures/tau_sweep_positions.pdf

================================================================================
TAU-SWEEP SUMMARY
================================================================================
     tau     BI mean     BI std     Success   Succ std   Seeds   Positions
  ------  ----------  ---------  ----------  ---------  ------  ----------
      50      0.0000     0.0000    0.004280   0.000000       1           1
     700      0.0000     0.0000    0.004600   0.000000       1           1

  Monotonicity check: PASSED
  Random guess: 0.003906

  Results CSV:   results/locality_diagnostics/tau_sweep_results.csv
```

```
  Summary CSV:   results/locality_diagnostics/tau_sweep_summary.csv
  Figures:       results/locality_diagnostics/figures

======================================================================
Done.
======================================================================
[exit code 0, 6 s]
```

```
$ cat results/locality_diagnostics/tau_sweep_results.csv && \
    ls results/locality_diagnostics/figures
tau,seed,best_position,profiling_accuracy,holdout_success,margin,certified_success,certified_advantage,bi_bi
    ts,n_holdout,n_profiling,n_positions_tested
50,0,325,0.006000,0.004280,0.016623,-0.012343,-0.016249,0.000000,25000,35000,1
700,0,0,0.005000,0.004600,0.016623,-0.012023,-0.015929,0.000000,25000,35000,1
tau_sweep_bi.pdf
tau_sweep_bi.png
tau_sweep_positions.pdf
tau_sweep_positions.png
tau_sweep_success.pdf
tau_sweep_success.png
[exit code 0, 0 s]
```

What to look for

- One row per window length and seed: where the window was placed, the success rate inside the window, how many traces were tested, and the success rate on the full trace for comparison.
- The script mixes training and test traces when it builds its test set, so the success rate is not a clean out-of-sample number. The paper presents this figure as a diagnostic, not as a certificate.